TraceSmith — GPU Profiling & Replay Architecture

Applications, pybind11 binding, C++17 core, GPU backends, and output formats

Dashed frame = architecture zone · Solid frame = subsystem · Box = module and responsibility

PROJECT OVERVIEW

TraceSmith is a cross-platform GPU profiling and deterministic replay system for AI compiler, framework, and driver engineers.
Its architecture separates applications, the pybind11 binding, a C++17 core, GPU backends, and output formats.

APPLICATION · user entry

- **C++ Application** — in-process capture
- **Python Application** — in-process capture
- **C++ CLI** — tracesmith profile · record · view
- **Python CLI** — tracesmith-cli export · analyze · replay
- **External UI** — Perfetto UI Chrome tracing

BINDINGS · pybind11 boundary

- **_tracesmith.so** — python/src/bindings.cpp — pybind11 native extension

CORE LIBRARY · C++17

Cluster Module

- **GPUTopology** — NVLink · PCIe · NVML
- **MultiGPUProfiler** — cross-device capture
- **TimeSync** — NTP / PTP / CUDA
- **NCCLTracker** — collective ops
- **ClockCorrelator** — clock alignment
- **CommAnalysis** — communication matrix

Capture Module

- **CUPTIProfiler** — NVIDIA CUDA
- **MCPTIProfiler** — MetaX MACA
- **MetalProfiler** — Apple Metal
- **MemoryProfiler** — GPU memory
- **AscendProfiler** — Huawei CANN
- **createProfiler()** — nsys · mcTracer xctrace · msprof

State Module

- **GPUStateMachine** — state analysis
- **InstructionStream** — dependency graph
- **TimelineBuilder** — ordered events
- **TimelineViewer** — ASCII rendering
- **PerfettoExporter** — JSON
- **PerfettoProtoExporter** — Protobuf